# Donovan Thach

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## Education

**Oregon State University** Corvallis, OR

*Bachelor of Science in Computer Science* *Sep. 2024 – June 2028*

- **GPA:** 3.63 | **Focus:** AI / ML
- **MECOP:** Selected for OSU's Multiple Engineering Cooperative Program, which places students in two paid 6-month industry co-ops, beginning Spring 2027
- **Honors:** Finley Academic Excellence Scholarship; OSU Honor Roll (5 terms)
- **Relevant Courses:** Data Structures, Algorithms, Operating Systems, Software Engineering, Computer Architecture, Web Development, Linear Algebra II, Calculus III; in progress: Databases, Machine Learning

## Experience

**Handshake AI Fellowship** Feb. 2026 – Present

*AI Model Evaluation Fellow* *Remote*

- Evaluated 500+ AI model outputs across multimodal tasks, including audio-visual retrieval and video understanding, through pairwise ranking and written rationales
- Annotated video with a hierarchical (L1/L2/L3) action taxonomy, producing labeled preference data used by external AI labs to fine-tune models

## Projects

**Oral Disease Analysis Model** | *Python, PyTorch, YOLO11, SLURM, ONNX* | GitHub Nov. 2025 – Present

- Built an end-to-end object detection pipeline (YOLO11, PyTorch) for 6 oral disease classes, merging 8 public datasets (∼6,100 photos) through a 4-stage remap, merge, split, and verify process
- Identified train/test leakage from duplicated augmented images that inflated mAP@50 to 81%; regrouped splits by source photo and added an automated leak check and locked test-set manifest, establishing a trustworthy 67.4% mAP@50 baseline
- Traced a 20-point validation/test gap in one class to misplaced and missing labels with a per-source audit script
- Ran training and evaluation as SLURM batch jobs on OSU's HPC cluster and exported models to ONNX

**NavigAid: AI Smart Cane** | *C++, Arduino, Python, YOLOv8, MiDaS* | GitHub May 2026

- Built an edge AI obstacle-avoidance cane for the visually impaired on a 3-person team, placing 2nd in the BeaverHacks 2026 Hardware Track; fused YOLOv8n and MiDaS depth with ultrasonic, laser, and gyroscope sensors, prioritizing ultrasonic readings within 40 cm
- Wrote the Arduino motor-steering firmware and serial link to the Raspberry Pi; wired most of the hardware
- Kept laser and gyroscope sensing in the final build despite a plan to cut it, reworking the wiring 8 hours before the deadline with ultrasonic sensors as a fallback

**WiFi Heart Rate & Fall Monitor** | *ESP32-S3, PyTorch, FastAPI, Snowflake, React* | GitHub May 2026

- Built a WiFi-only, camera-free elderly monitoring system on a 3-person team, streaming Channel State Information from two ESP32-S3 boards into a FastAPI → Snowflake → React pipeline
- Trained the team's PyTorch CNN fall classifier and LSTM heart and breathing rate estimator on CSI data, led on-site collection of labeled fall and vitals sessions, and streamed live CSI from both boards

**Lift Log** | *React, TypeScript, Express, MongoDB* Jan. 2026 – Mar. 2026

- Built a full-stack workout planner on a 7-person team in 1–2 week sprints, earning the top score in the course
- Built JWT cookie authentication with route-guard middleware and lockout after 5 failed logins; wrote team UATs

## Technical Skills

**Languages**: Python, C/C++, SQL, JavaScript, TypeScript, x86-64 Assembly, HTML/CSS
**Frameworks**: React, React Native, Flask, Node.js, FastAPI
**Developer Tools**: Git, Linux, SLURM, Vite, npm
**Libraries & APIs**: PyTorch, YOLO11, ONNX, NumPy, pandas, Snowflake, Firebase, OpenAI API, Three.js